

remain responsive and beneficial to all participants. This approach helps us create a comprehensive and cutting-edge curriculum.

How will the centre keep up with technological advancements in the field?

We plan to start with state-of-the-art equipment and a flexible lab design that will adapt to tech advancements. Initially, equipment purchases will align with industry needs. We are fully aware of the dynamic nature of the electronics manufacturing industry and the ongoing advancements, such as integrating automated robots in pick-and-place (PNP) machines and including cleanroom facilities. To address this, we will design the lab in a way that allows for flexibility and scalability. Our focus is on teaching fundamental manufacturing processes, and we plan to update equipment and programmes as needed to stay relevant.

How does the centre enable innovation, research, and materials development?

The centre will focus on advancing materials and processes for electronic manufacturing, enhancing efficiency, reliability, and sustainability. My interest in materials development for SMT applications will guide efforts to improve overall performance and environmental impact. The research will cover reflow ovens, solder paste printers, and automated inspections, focusing on developing new materials for SMT applications and exploring structure-property correlations. For example, there is a strong drive in the industry towards low-temperature solder development to enable soldering of packages with aggressive form factors, reducing carbon-dioxide emissions and power consumption.

How will this centre enable electronics startups?

IIT Kanpur already has a very strong incubation ecosystem. In fact, in terms of innovation, it was ranked

first last year in the NIRF Innovations rankings. We have a robust incubator called SIIC IIT Kanpur, which houses several companies focusing on electronics manufacturing. We also have one of the largest research and technology parks in India, Technopark, that houses facilities for several companies working with IIT Kanpur faculties on different R&D problems. This new centre will serve as a support system for these companies as well, providing them with access to state-of-the-art facilities and training programmes.

What is NMTronics' role, and who will maintain the lab?

NMTronics plays a crucial role in funding the initiative through CSR contributions. The lab is owned and managed fully by IIT Kanpur. The work is done with guidance from administrative and technical boards and industry experts. Project-based staff hired through the university will handle day-to-day maintenance, ensuring the centre's smooth operation. Industry professionals will contribute to training and curriculum development, enhancing the lab's effectiveness.

How will the equipment procurement process work?

NMTronics contributes CSR funding. Equipment procurement will adhere to government norms through the Government e-Marketplace (GeM), ensuring transparency and preventing conflicts of interest. This approach allows us to source the best equipment from various suppliers.

If any other firm wants to donate equipment, can they, and who should they contact?

Any individual or organisation interested in contributing in any manner may contact us. We welcome contributions from everyone aligned with our mission of growing and supporting the electronics manufacturing sector in India.

Will there be any certification or

accreditation for the programmes offered at the centre?

Yes, we plan to provide Joint Certification & Diploma to be issued jointly by NMTronics and IIT Kanpur for the training courses offered at the centre. This will help participants demonstrate their acquired skills and knowledge in electronics manufacturing and SMT processes. In fact, we aim to provide certified trainees with internships and potential job opportunities with industry partners, benefiting both trainees and the industry. The centre will be open to academic institutions and industry partners, offering a seamless transition from certification to internships to full-time employment.

Does that mean that firms/startups outside IIT Kanpur can also get their employees trained here?

Yes, definitely. This centre is open for startups for building NPI along with all electronics manufacturers to use as R&D and process consultancy.

What are the costs associated with enrolling for such trainings?

The cost of the training programs is being finalised. The pricing will be determined by the technical and administrative advisory boards. The endeavour is to establish a fair fee structure that reflects the value of the training provided.

Is there a maximum capacity for training or accessing the centre at any given time?

We have not set a maximum capacity yet. Similar centres accommodate 25 to 50 people per batch, but our model may differ based on facility size and the nature of training. Details are still being worked out.

By when will the setup be ready? And, when is the first batch expected to start?

Work is in progress, and the lab is expected to be fully functional by August 2024, and that is when we plan to initiate our first batch. **EFY**